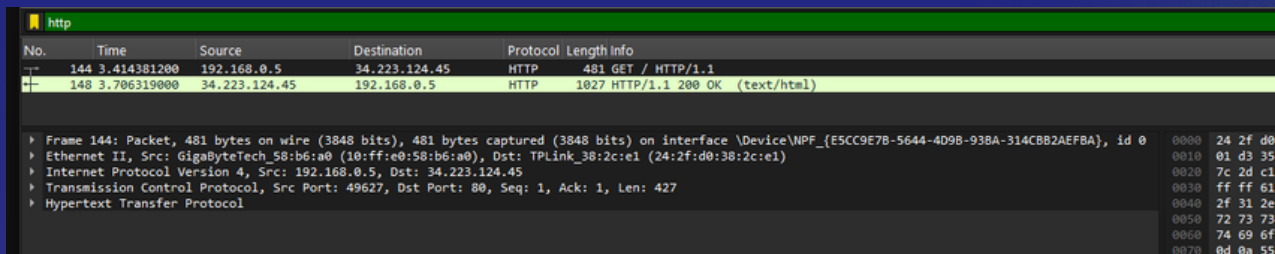


Lab1

PART 1: START WIRESHARK AND CAPTURE PACKETS & FILTER HTTP PACKETS AND ANALYZE THEM.



No.	Time	Source	Destination	Protocol	Length	Info
144	3.414381200	192.168.0.5	34.223.124.45	HTTP	481	GET / HTTP/1.1
148	3.706319000	34.223.124.45	192.168.0.5	HTTP	1027	HTTP/1.1 200 OK (text/html)

Frame 144: Packet, 481 bytes on wire (3848 bits), 481 bytes captured (3848 bits) on interface \Device\NPF_{E5CC9E7B-5644-4D9B-93BA-314CBB2AEFBA}, id 0

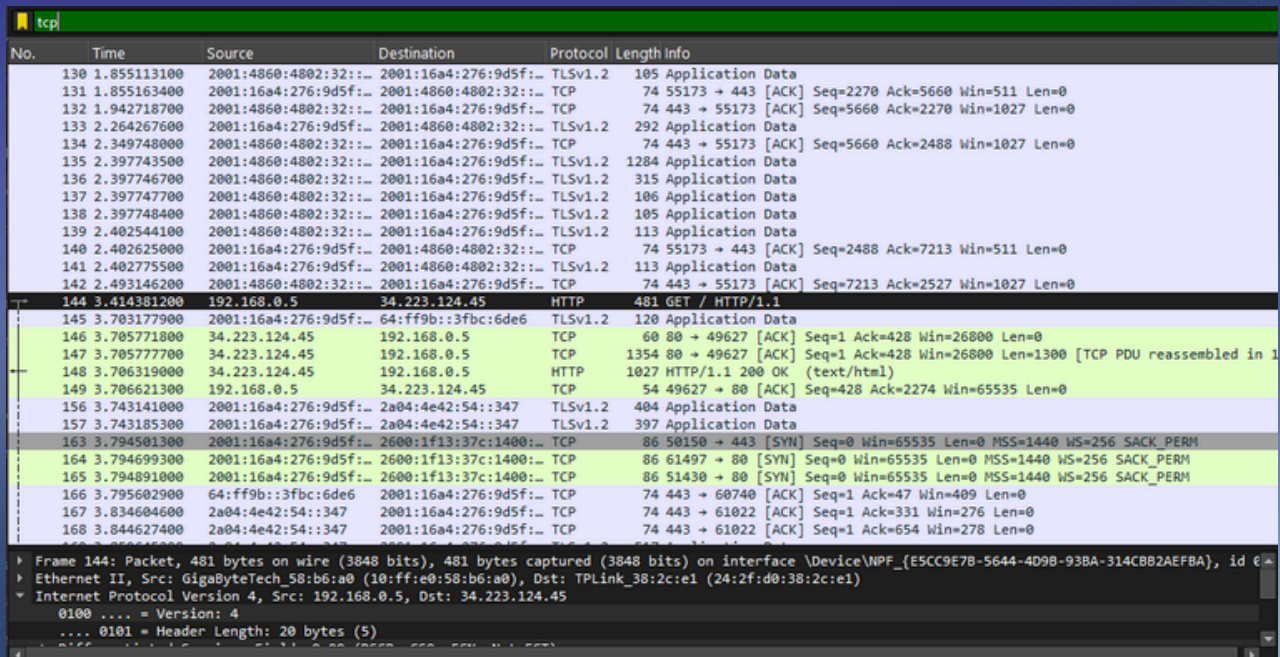
Ethernet II, Src: GigaByteTech_58:b6:a0 (10:ff:e0:58:b6:a0), Dst: TPLink_38:2c:e1 (24:2f:d0:38:2c:e1)

Internet Protocol Version 4, Src: 192.168.0.5, Dst: 34.223.124.45

Transmission Control Protocol, Src Port: 49627, Dst Port: 80, Seq: 1, Ack: 1, Len: 427

Hypertext Transfer Protocol

PART 2: FILTER TCP PACKETS



No.	Time	Source	Destination	Protocol	Length	Info
130	1.855113100	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	105	Application Data
131	1.855163400	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TCP	74	55173 → 443 [ACK] Seq=2270 Ack=5660 Win=511 Len=0
132	1.942718700	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TCP	74	443 → 55173 [ACK] Seq=5660 Ack=2270 Win=1027 Len=0
133	2.264267600	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	292	Application Data
134	2.349748000	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TCP	74	443 → 55173 [ACK] Seq=5660 Ack=2488 Win=1027 Len=0
135	2.397743500	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	1284	Application Data
136	2.397746700	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	315	Application Data
137	2.397747700	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	106	Application Data
138	2.397748400	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	105	Application Data
139	2.402544100	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	113	Application Data
140	2.402625000	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TCP	74	55173 → 443 [ACK] Seq=2488 Ack=7213 Win=511 Len=0
141	2.402775500	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TLSv1.2	113	Application Data
142	2.493146200	2001:16a4:276:9d5f::...	2001:16a4:276:9d5f::...	TCP	74	443 → 55173 [ACK] Seq=7213 Ack=2527 Win=1027 Len=0
144	3.414381200	192.168.0.5	34.223.124.45	HTTP	481	GET / HTTP/1.1
145	3.703177900	2001:16a4:276:9d5f::...	64:ff9b::3fbc:6de6	TLSv1.2	120	Application Data
146	3.705771800	34.223.124.45	192.168.0.5	TCP	60	80 → 49627 [ACK] Seq=1 Ack=428 Win=26800 Len=0
147	3.705777700	34.223.124.45	192.168.0.5	TCP	1354	80 → 49627 [ACK] Seq=1 Ack=428 Win=26800 Len=1300 [TCP PDU reassembled in 1]
148	3.706319000	34.223.124.45	192.168.0.5	HTTP	1027	HTTP/1.1 200 OK (text/html)
149	3.706621300	192.168.0.5	34.223.124.45	TCP	54	49627 → 80 [ACK] Seq=428 Ack=2274 Win=65535 Len=0
156	3.743141000	2001:16a4:276:9d5f::...	2a04:4e42:54::347	TLSv1.2	404	Application Data
157	3.743185300	2001:16a4:276:9d5f::...	2a04:4e42:54::347	TLSv1.2	397	Application Data
163	3.794501300	2001:16a4:276:9d5f::...	2600:1f13:37c:1400::...	TCP	86	50150 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
164	3.794699300	2001:16a4:276:9d5f::...	2600:1f13:37c:1400::...	TCP	86	61497 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
165	3.794891000	2001:16a4:276:9d5f::...	2600:1f13:37c:1400::...	TCP	86	51430 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
166	3.795602900	64:ff9b::3fbc:6de6	2001:16a4:276:9d5f::...	TCP	74	443 → 60740 [ACK] Seq=1 Ack=47 Win=409 Len=0
167	3.834604600	2a04:4e42:54::347	2001:16a4:276:9d5f::...	TCP	74	443 → 61022 [ACK] Seq=1 Ack=331 Win=276 Len=0
168	3.844627400	2a04:4e42:54::347	2001:16a4:276:9d5f::...	TCP	74	443 → 61022 [ACK] Seq=1 Ack=654 Win=278 Len=0

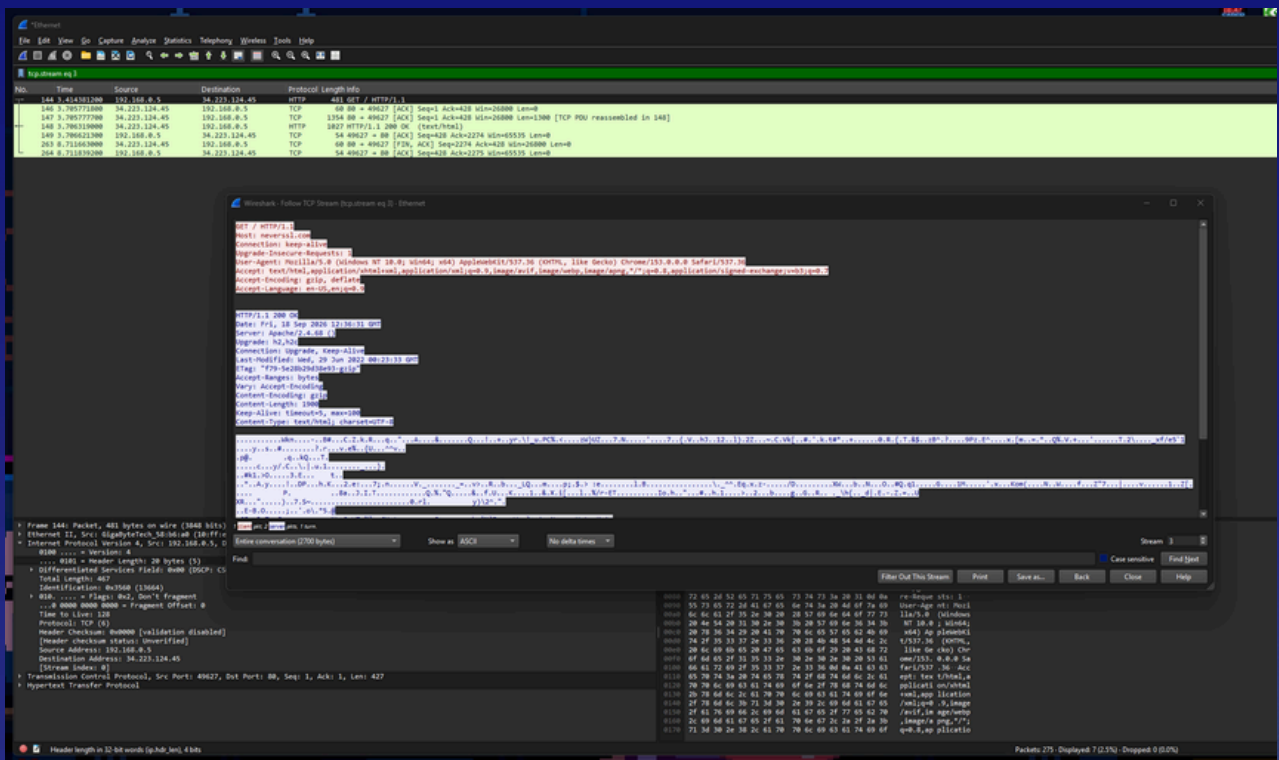
Frame 144: Packet, 481 bytes on wire (3848 bits), 481 bytes captured (3848 bits) on interface \Device\NPF_{E5CC9E7B-5644-4D9B-93BA-314CBB2AEFBA}, id 0

Ethernet II, Src: GigaByteTech_58:b6:a0 (10:ff:e0:58:b6:a0), Dst: TPLink_38:2c:e1 (24:2f:d0:38:2c:e1)

Internet Protocol Version 4, Src: 192.168.0.5, Dst: 34.223.124.45

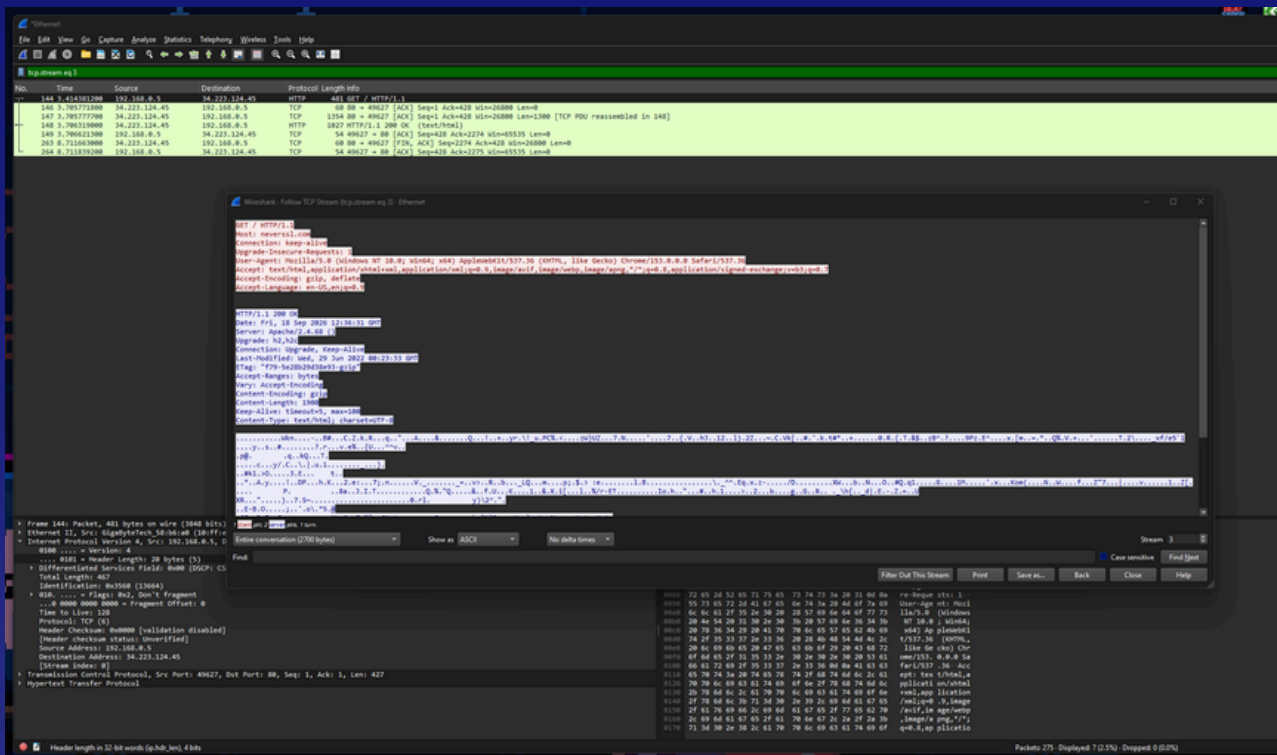
0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)



PART 2:ANALYZE TCP HANDSHAKE AND INVESTIGATE DATA TRANSFER AND TERMINATION

144	3.414381200	192.168.0.5	34.223.124.45	HTTP	481 GET / HTTP/1.1
145	3.705777900	2001:16a4:276:9d5f::64	ff9b::3fbc:6de6	TLSv1.2	120 Application Data
146	3.705777800	34.223.124.45	192.168.0.5	TCP	60 80 → 49627 [ACK] Seq=1 Ack=428 Win=26800 Len=0
147	3.705777700	34.223.124.45	192.168.0.5	TCP	1354 80 → 49627 [ACK] Seq=1 Ack=428 Win=26800 Len=1300 [TCP PDU reassembled in 148]
148	3.706319000	34.223.124.45	192.168.0.5	HTTP	1027 HTTP/1.1 200 OK (text/html)
149	3.706621300	192.168.0.5	34.223.124.45	TCP	54 49627 → 80 [ACK] Seq=428 Ack=2274 Win=65535 Len=0
156	3.743141000	2001:16a4:276:9d5f::2a04	4e42:54:347	TLSv1.2	404 Application Data
157	3.743105300	2001:16a4:276:9d5f::2a04	4e42:54:347	TLSv1.2	397 Application Data
163	3.794501300	2001:16a4:276:9d5f::2600	1f13:37c:1400::	TCP	86 50150 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
164	3.794699300	2001:16a4:276:9d5f::2600	1f13:37c:1400::	TCP	86 51497 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
165	3.794891000	2001:16a4:276:9d5f::2600	1f13:37c:1400::	TCP	86 51430 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
166	3.795402900	64:ff9b::3fbc:6de6	2001:16a4:276:9d5f::	TCP	74 443 → 60740 [ACK] Seq=1 Ack=427 Win=409 Len=0
167	3.834604600	2a04:4e42:54:347	2001:16a4:276:9d5f::	TCP	74 443 → 61022 [ACK] Seq=1 Ack=331 Win=276 Len=0
168	3.844627400	2a04:4e42:54:347	2001:16a4:276:9d5f::	TCP	74 443 → 61022 [ACK] Seq=1 Ack=654 Win=278 Len=0
169	3.859645200	2a04:4e42:54:347	2001:16a4:276:9d5f::	TLSv1.2	517 Application Data
170	3.883037400	2a04:4e42:54:347	2001:16a4:276:9d5f::	TLSv1.2	109 Application Data
171	3.883043100	2a04:4e42:54:347	2001:16a4:276:9d5f::	TLSv1.2	481 Application Data
172	3.883148200	2001:16a4:276:9d5f::	2a04:4e42:54:347	TCP	74 61022 → 443 [ACK] Seq=654 Ack=886 Win=255 Len=0
175	4.035893900	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	86 443 → 50150 [ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 SACK_PERM WS=4096
176	4.035973400	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	74 50150 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
177	4.037104300	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	86 51431 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
262	8.607915500	192.168.0.5	162.159.135.234	TCP	54 64983 → 443 [ACK] Seq=122 Ack=36 Win=253 Len=0
263	8.711663000	34.223.124.45	192.168.0.5	TCP	60 80 → 49627 [FIN, ACK] Seq=2274 Ack=428 Win=26800 Len=0



PART 2:ANALYZE TCP HANDSHAKE AND INVESTIGATE DATA TRANSFER AND TERMINATION

No.	Time	Source	Destination	Protocol	Length	Info
61472	26.243614700	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	86	60565 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
68386	27.243925100	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	86	[TCP Retransmission] 60565 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
68679	29.248971200	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	86	[TCP Retransmission] 60565 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
77997	33.255285000	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	86	[TCP Retransmission] 60565 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
78452	33.478624900	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	78	80 → 60565 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1400
78453	33.478950700	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	74	60565 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
78454	33.483765200	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	HTTP	592	GET /online HTTP/1.1
78928	33.855675200	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	74	80 → 60565 [ACK] Seq=1 Ack=519 Win=26800 Len=0
79141	33.857370500	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	HTTP	663	HTTP/1.1 301 Moved Permanently (text/html)
79214	33.864236400	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	HTTP	593	GET /online/ HTTP/1.1
80107	34.107265100	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	74	80 → 60565 [ACK] Seq=590 Ack=1038 Win=27872 Len=0
80109	34.107290400	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	1374	80 → 60565 [ACK] Seq=590 Ack=1038 Win=27872 Len=1300 [TCP PDU reassembled in 80110]
80110	34.107291200	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	HTTP	293	HTTP/1.1 200 OK (text/html)
80198	34.139986800	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	HTTP	505	GET /favicon.ico HTTP/1.1
80512	34.376765500	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	HTTP	490	HTTP/1.1 200 OK (PNG)
80559	34.418308300	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	74	60565 → 80 [ACK] Seq=1469 Ack=2525 Win=65119 Len=0
92430	39.378303900	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	74	80 → 60565 [FIN, ACK] Seq=2525 Ack=1469 Win=28944 Len=0
92431	39.378615300	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	74	60565 → 80 [ACK] Seq=1469 Ack=2526 Win=65119 Len=0
97342	41.922096200	2001:16a4:276:9d5f::	2600:1f13:37c:1400::	TCP	74	60565 → 80 [FIN, ACK] Seq=1469 Ack=2526 Win=65119 Len=0
98122	42.156095900	2600:1f13:37c:1400::	2001:16a4:276:9d5f::	TCP	74	80 → 60565 [ACK] Seq=2526 Ack=1470 Win=28944 Len=0

PART 3:: GENERATE UDP TRAFFIC AND CAPTURE PACKETS & FILTER AND ANALYSIS UDP PACKETS

udp					
No.	Time	Source	Destination	Protocol	Length Info
2	0.657885300	2001:16a4:276:9d5f::	2001:4860:482d:400::	UDP	1285 58165 → 443 Len=1223
3	0.657846200	2001:16a4:276:9d5f::	2001:4860:482d:400::	UDP	1292 58165 → 443 Len=1230
4	0.657859400	2001:16a4:276:9d5f::	2001:4860:482d:400::	UDP	1292 58165 → 443 Len=1230
5	0.657874500	2001:16a4:276:9d5f::	2001:4860:482d:400::	UDP	656 58165 → 443 Len=594
7	0.755941800	2001:4860:482d:400::	2001:16a4:276:9d5f::	UDP	92 443 → 58165 Len=30
8	0.787366100	2001:16a4:276:9d5f::	2001:4860:482d:400::	UDP	95 58165 → 443 Len=33

▼ Frame 3: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{E5CC9E7B-5644-4D9B-93BA-314CBB2AEFBA}, id 6
Section number: 1

- ▶ Interface id: 0 (\Device\NPF_{E5CC9E7B-5644-4D9B-93BA-314CBB2AEFBA})
- Encapsulation type: Ethernet (1)
- Arrival Time: Sep 18, 2026 16:26:41.662810200 Arab Standard Time
- UTC Arrival Time: Sep 18, 2026 13:26:41.662810200 UTC
- Epoch Arrival Time: 1789738001.662810200
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 40.900 microseconds]
- [Time delta from previous displayed frame: 40.900 microseconds]
- [Time since reference or first frame: 657.846200 milliseconds]
- Frame Number: 3
- Frame Length: 1292 bytes (10336 bits)
- Capture Length: 1292 bytes (10336 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ipv6:udp:data]
- Character encoding: ASCII (0)
- [Coloring Rule Name: UDP]
- [Coloring Rule String: udp]

▼ Ethernet II, Src: GigaByteTech_58:b6:a0 (10:ff:e0:58:b6:a0), Dst: TPLink_38:2c:e1 (24:2f:d0:38:2c:e1)

- ▶ Destination: TPLink_38:2c:e1 (24:2f:d0:38:2c:e1)
- ▶ Source: GigaByteTech_58:b6:a0 (10:ff:e0:58:b6:a0)
- Type: IPv6 (0x86dd)
- [Stream index: 1]

▼ Internet Protocol Version 6, Src: 2001:16a4:276:9d5f:7dc3:c3dd:6497:99b0, Dst: 2001:4860:482d:400::

0110 = Version: 6

- ▶ 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
- ▶ 1000 0100 1001 0111 1101 = Flow Label: 0x8497d
- Payload Length: 1238
- Next Header: UDP (17)
- Hop Limit: 64
- ▶ Source Address: 2001:16a4:276:9d5f:7dc3:c3dd:6497:99b0
- ▶ Destination Address: 2001:4860:482d:400::
- [Stream index: 1]

▼ User Datagram Protocol, Src Port: 58165, Dst Port: 443

- Source Port: 58165
- Destination Port: 443
- Length: 1238
- Checksum: 0xcfd9 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 0]
- [Stream Packet Number: 2]
- ▶ [Timestamps]
- UDP payload (1230 bytes)

▼ Data (1230 bytes)

Data [...]: 52f6249f187a9bdc19cf15162b181f5e6e57138ef0b644a7667a40e3ff23598beac2edf46a9ae79a325800c3b007c6c65159d15c3f9ff5ef52d1aa59525e3aa83372f60559...

[Length: 1230]

PART 4:

TASK 1

	TCP OR UDP	REASONS
RELIABILITY AND CONNECTION ESTABLISHMENT	TCP	TCP uses a three-way handshake (SYN, SYN-ACK, ACK) to establish a connection before any data is sent, and it guarantees delivery through acknowledgments and retransmissions. UDP is connectionless: it sends datagrams with no handshake and no delivery guarantee.
DATA INTEGRITY AND ORDERING	TCP	TCP uses sequence numbers to reorder packets that arrive out of order and checksums plus retransmission to recover lost or corrupted data, ensuring an accurate, in-order byte stream. UDP only has a basic checksum and provides no reordering or retransmission, so packets can arrive out of order, be duplicated, or be lost.

TASK 2

	TCP	UDP
USE CASES	Web browsing (HTTP/HTTPS), email (SMTP/IMAP), file transfer (FTP), remote login (SSH) - any application where complete, ordered, error-free data delivery matters more than speed.	Video/audio streaming, VoIP calls, online gaming, DNS lookups, live broadcasts - applications that tolerate some data loss but require low latency and real-time performance.
PERFOR MANCE	Slower due to connection setup, acknowledgments, flow control, and error correction overhead, but highly reliable and accurate.	Faster and more efficient with minimal overhead (smaller header, no handshake), but offers no guarantee that data arrives, arrives in order, or arrives at all.